

Quantum Machine Learning for Early Disease Prediction

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ABSTRACT

Quantum Machine Learning (QML) is an emerging approach that combines quantum computing with machine learning to explore new methods for analyzing and classifying complex data. In healthcare, QML can be applied to patient-related information to identify patterns associated with different diseases. This study focuses on the application of QML for heart disease prediction using the UCI Heart Disease dataset. The proposed approach combines classical data preprocessing, feature scaling, and Principal Component Analysis (PCA) with quantum feature mapping and quantum-kernel-based classification. A four-qubit ZZFeatureMap and Fidelity Quantum Kernel are used with a Quantum Support Vector Classifier (QSVC), while a classical Support Vector Classifier (SVC) is used as a baseline for comparison. The models are evaluated using accuracy, precision, recall, F1-score, and a confusion matrix. The study demonstrates a practical hybrid classical–quantum workflow for heart disease classification and highlights the current limitations of QML, including limited dataset size, feature reduction, and the computational cost of quantum-kernel simulation. The work provides a foundation for further research into scalable and clinically applicable QML-based healthcare systems.

Index Terms— Quantum Machine Learning, Quantum Computing, Heart Disease Prediction, QSVC, Fidelity Quantum Kernel, PCA, Support Vector Machine, Healthcare.

INTRODUCTION

Early disease prediction is important in healthcare because identifying disease-related risks at an early stage can support timely clinical attention and better patient management. Machine learning techniques are already widely used to analyze healthcare data and perform disease classification. However, healthcare datasets may contain different types of features, varying scales, and complex relationships, which require suitable preprocessing and representation before classification.

Quantum Machine Learning provides an emerging approach by combining quantum computing principles with machine learning methods. QML techniques such as quantum kernels, Quantum Support Vector Machines, and quantum neural networks have been investigated for classification and prediction tasks. These approaches are particularly useful as research tools for studying how quantum feature representations can be incorporated into conventional machine-learning workflows.

In this study, a hybrid classical–quantum framework is developed for heart disease classification using the UCI Heart Disease dataset. The workflow performs data preprocessing and feature scaling, followed by Principal Component Analysis (PCA) to reduce the original 13 clinical features to four components. The reduced features are encoded using a four-qubit ZZFeatureMap, and a Fidelity Quantum Kernel is used with QSVC for classification. A classical SVC is evaluated using the same matched training subset and test set to provide a controlled comparison. The study evaluates the practical use of a quantum-kernel-based approach while considering the limitations of simulation, dataset size, feature reduction, and current quantum computing resources.